

DRISHTI AI HACKATHON 2026

VIGILENSE

- Problem Statement ID - 2
- Problem Statement Title - Offline Video Segmentation and ROI Detection Using Motion Estimation
- Team ID - DI2_40
- Team Name - Mackintosh

OUR APPROACH

PROPOSED SOLUTION

- **GPU Video Pipeline** – DeepStream **NVDEC** streams footage into **NVOF** 4×4 optical flow without loading complete recordings into RAM.
- **Seat Level Segmentation** – **Calliberated** desk polygons use median motion baselines and hysteresis to generate stable event boundaries.
- **Multi-Evidence Fusion** – Motion, object confidence, pose deviation, duration, repetition, and neighbouring-seat interaction jointly rank events.
- **Tiny Object Detection** – **Objects365** trained **D-FINE**-L analyses overlapping full-resolution seat crops with cross-frame phone-chit stationery disambiguation.
- **Searchable Evidence Timeline** – Generates timestamped clips, heatmaps, overlays, reviewer feedback and event logs.
- **Crowded Pose Tracking & VLM Verif.** – **RTMO** keypoints, **NvDCF** trajectories, and seat reconciliation maintain attribution while **Gemma 4 E4B** QAT-Q4 evaluates even further and outputs evidence with low RAM.

Explainable Evidence Prioritization

Produces confidence-ranked, uncertainty-aware events with complete provenance, enabling efficient human investigation without automated misconduct decisions.

Hardware-Accelerated Motion Gating

Leverages NVIDIA Optical Flow (NVOF) to generate candidate events before invoking computationally expensive AI models.

Temporal Multi-Evidence Fusion

Integrates motion, object, pose, tracking, and temporal consistency into a unified evidence score instead of relying on single-frame predictions.

Calibration-Aware Spatial Localization

Uses calibrated seat and desk ROIs with global motion compensation to achieve robust event attribution in crowded examination halls.

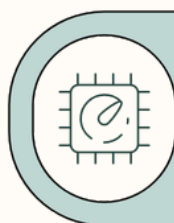
High-Resolution ROI Inference

Performs tiled, high-resolution inference on candidate seat regions to improve detection of small prohibited objects such as phones and chits.

Multi-Stage Evidence Cascade

Executes detection, tracking, pose estimation, and visual verification only on candidate clips, minimizing false positives and computational overhead.

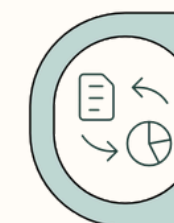
UNIQUE VALUE PROPOSITION



Hardware-Aware Pipeline
 DeepStream and NVIDIA Optical Flow filter candidate events before running expensive AI models.



Evidence-Driven Engine
 Motion, object detection, pose, tracking, and temporal reasoning are fused before event prioritization.



Explainable AI
 Every event includes confidence, uncertainty, supporting evidence, and full audit provenance.



Seat-Anchored Intelligence
 Calibrated seat and desk ROIs ensure accurate event localization without relying on biometric identity.



Adaptive Resource Scheduling
 Heavyweight models like Gemma are invoked only for ambiguous or high-priority evidence.



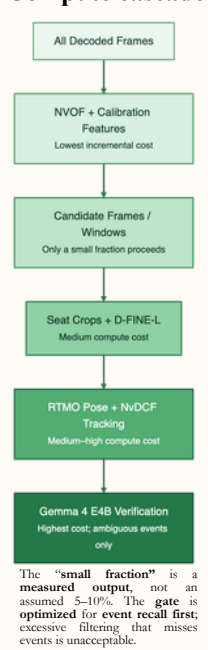
Investigation-Oriented Architecture
 Produces ranked, searchable evidence for human investigators instead of automated misconduct decisions.

Layer	Selected Technology	What It Does
Core Languages	Python, TS, C++ Pugins	CV, backend services, UI, and GPU extensions
Video Processing	DeepStream 8, GStreamer, NVDEC, FFmpeg	Zero-copy decoding, video validation, seeking, and clip generation
Motion & ROI	NVOF <u>gst-nvof</u> , OpenCV Homography, Seat Polygons	4×4 optical flow, camera compensation, and seat-level event generation
Object Detection	D-FINE-L, Objects365, TensorRT FP16, Supervision	Tiled high-resolution detection of phones, chits, and prohibited objects
Pose Estimation	MMPose RTMO	Multi-person head, wrist, and torso keypoint extraction
Tracking & Identity	NvDCF, Seat-Reconciliation Engine	Maintains seat attribution through overlap, occlusion, and track loss
Event Intelligence	Median/MAD, Hysteresis FSM, Evidence Fusion	Maintains seat attribution through overlap, occlusion, and track loss
VLM Verification	Gemma 4 E4B QAT-Q4, llama.cpp CUDA	Verifies 4–8 event frames with confidence, uncertainty, or abstention
Data & Interface	PostgreSQL, NVMe/NAS, React, ECharts	Stores evidence and presents clips, timelines, heatmaps, and search
Deployment	Docker Compose, NVIDIA Container Toolkit, NVML	Offline RTX 4070–4090 deployment with RAM, VRAM, and health monitoring

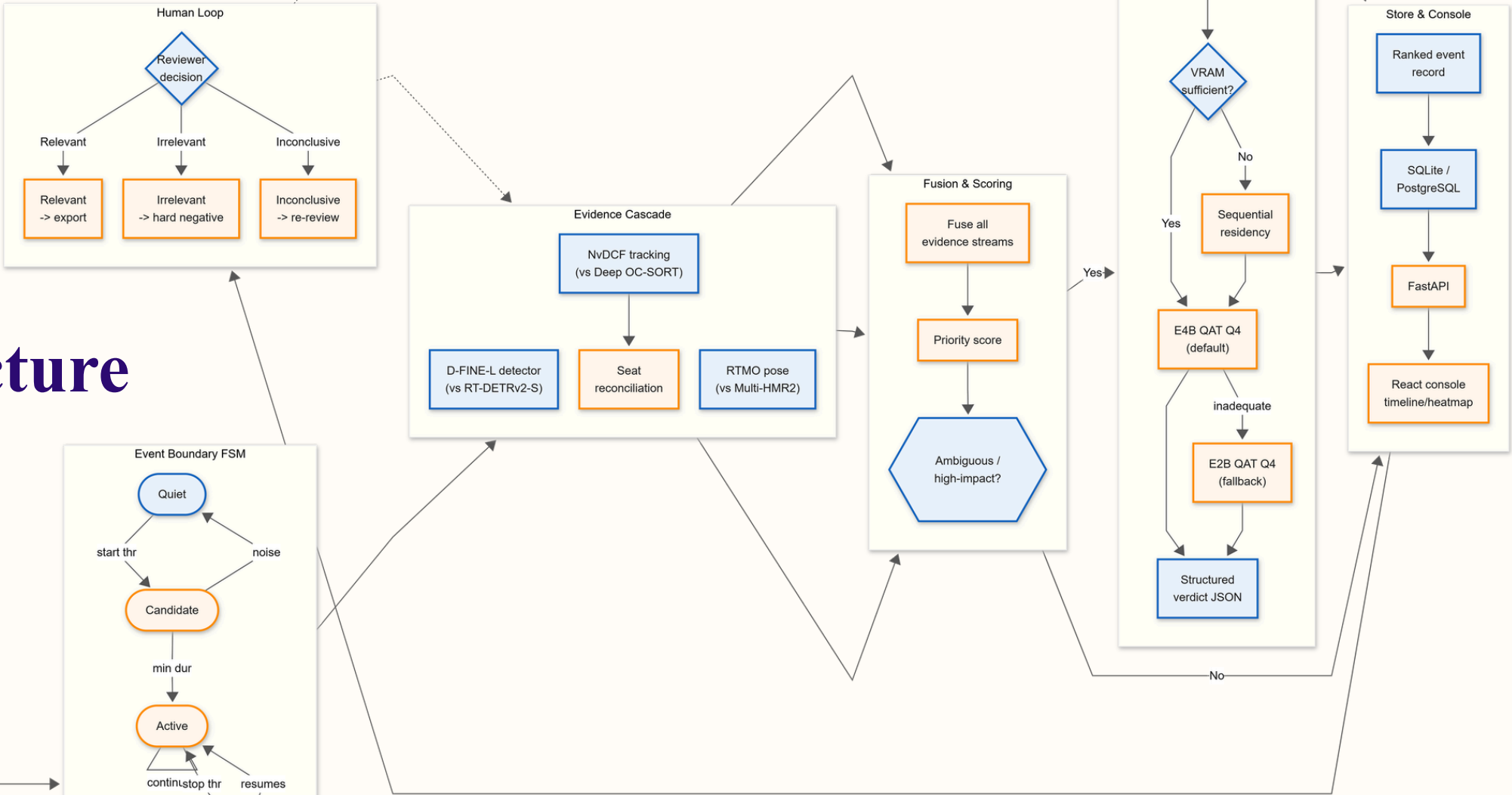
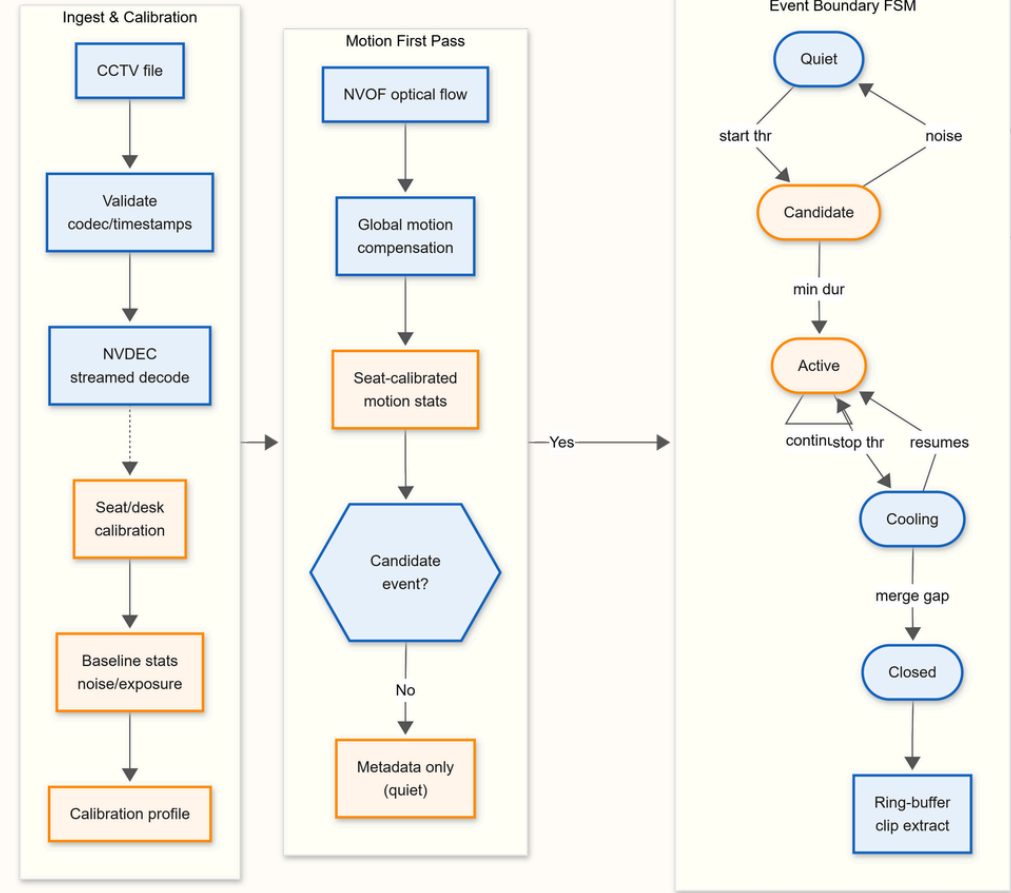
Offline Deployment: Docker Compose + NVIDIA Runtime · RTX 4070–4090 · No Cloud Video Transfer

TECHNICAL APPROACH

Compute cascade



System Architecture



KEY CAPABILITIES

- Offline AI
- Motion Detection
- ROI Extraction
- Pose Tracking
- Event Intelligence

FEASIBILITY AND VIABILITY

Challenges & Risks

Crowded Examination Halls
Overlapping students cause missing poses, incorrect ROIs, and ID switches.

Subtle Motion & Tiny Objects
Small gestures, phones, and chits may be missed or confused.

Lighting, Noise & Camera Motion
Shadows, compression, fans, or vibration create false motion ROIs.

False Alerts & Segmentation
Normal actions over-trigger; incidents become fragmented or incorrectly merged.

Duration, Scale & Failures
Multi-camera footage overloads compute, storage, or fails mid-processing.

Generalization, Privacy & Trust
New halls reduce accuracy; opaque surveillance creates security and ethical risks

Mitigation Strategy

RTMO + NvDCF + seat polygons restore reliable seat-level attribution.

D-FINE-L + tiled TensorRT crops + temporal confirmation recover small evidence.

NVOF + homography + MAD baselines suppress global and environmental changes.

Hysteresis FSM + duration windows + evidence fusion stabilize events

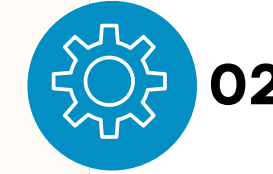
DeepStream/NVDEC + Redis/RQ + PostgreSQL/NAS checkpoints enable recovery and scale

Per-hall calibration + offline Docker + Gemma explanations + human review preserve trust.



01

Existing Infrastructure Fit
Works with recorded CCTV and cheap GPU systems; no camera replacement or cloud dependency.



02

Measurable Operational Value
Ranked clips and searchable events reduce hours of manual footage review



03

Controlled Running Cost
Motion gating runs the models. Only highly flagged images go to VLM.



04

Low Adoption Risk
No facial recognition; every alert includes evidence, uncertainty, and mandatory human review.



05

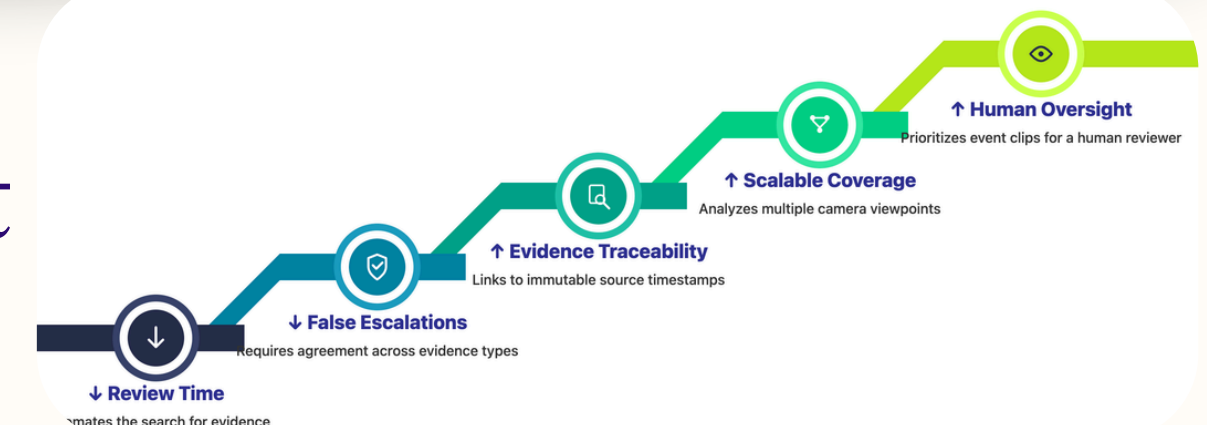
Scalable Deployment
Dockerized Redis/Cloudflare workers scale by video, camera, hall, or examination centre



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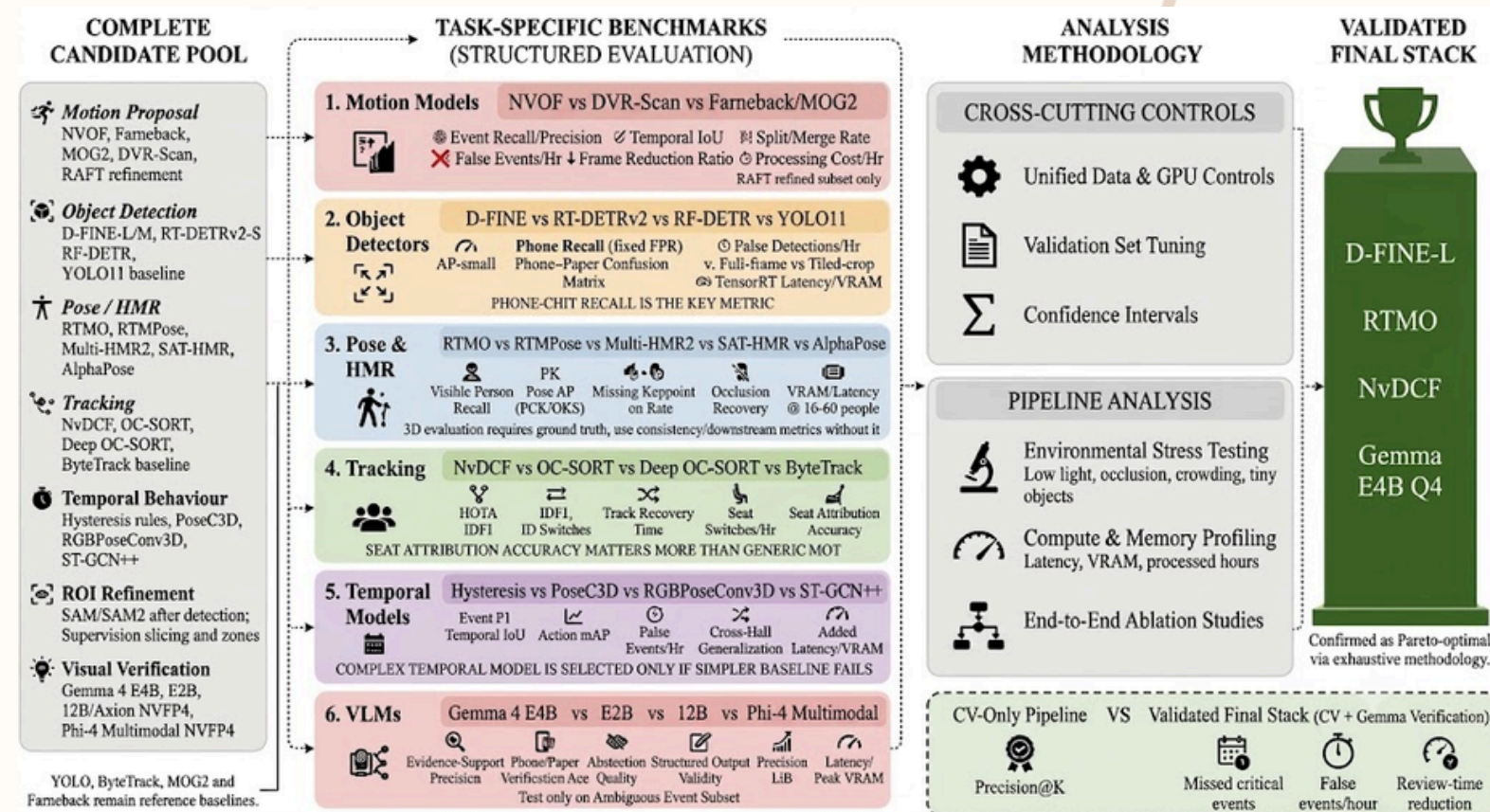
Sustainable Business Model
Supports easy licensing, classroom specific models, scalable infra, low cost, attractive CRMs

Impact



COMPUTATIONAL ANALYSIS & EVALS

Layer	Stack	Benchmark Metrics	Resource Budget	Edge Cases	Control / Decision
Motion & Events	DeepStream, NVDEC, NVOF, MAD-Hysteresis	Recall, t-IoU, FP/hour, split/merge rate	Lowest cost; processes all frames	Lighting, noise, fans, camera shake	Global compensation and adaptive seat baselines
Object Detection	D-FINE-L TensorRT; RT-DETRv2 challenger	AP-small, phone recall, paper confusion, latency	Candidate seat crops only	Tiny/hidden phones, paper, stationery	Tiled multi-frame detection or abstention
Pose & Tracking	RTMO, NvDCF; Deep OC-SORT challenger	PCK, IDF1, HOTA, ID switches	Selected clips; test 10/30/60 people	Occlusion, overlap, seat changes	Seat reconciliation and multi-seat uncertainty
VLM Verification	Gemma 4 E4B Q4; E2B fallback	Evidence precision, hallucination, abstention, VRAM	Sequential on 12 GB; 15–20% reserve	Hallucination, OOM, unclear footage	Structured output; cannot declare misconduct
System & Storage	RTX 4070-4090, 64-128 GB RAM, PostgreSQL, NAS	Throughput, peak memory, review reduction	3h/camera: 5.4–10.8 GB; 8 cams: 43–86 GB	Disk/power failure, sync, domain shift	Bounded queues, checkpoints, 20% free disk, human review



Candidates progress through **task** specific, cross cutting and e2e benchmarks; only the **Pareto Analysed** optimal stack meeting **accuracy**, false-alarm, **latency** and **VRAM constraints** is selected.

Computational Analysis - Selective GPU inference, bounded memory, and evidence-driven verification enable accurate multi-hour analysis on a single RTX workstation.

Edge Case Analysis

1

Low Illumination
Reliable detection under varying lighting and exposure conditions.

2

Occlusions
Maintains event tracking despite partial visibility of candidates.

3

Dense Seating
Accurate seat-level localization in crowded examination halls.

4

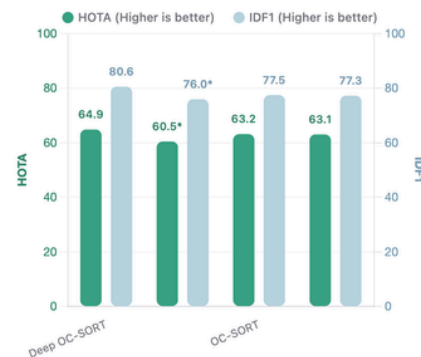
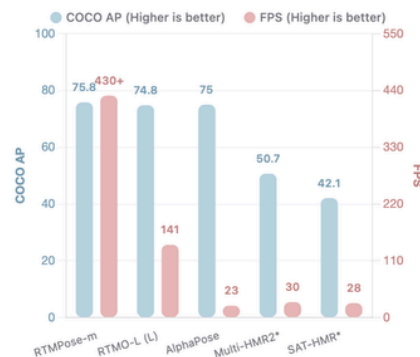
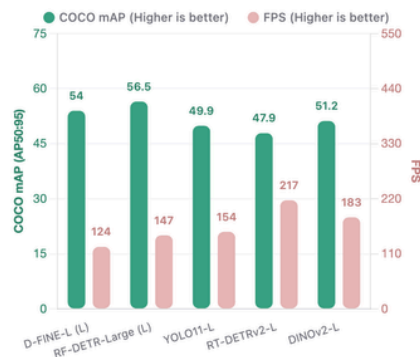
Camera Motion
Reduces false detections using global motion compensation.

5

Small Object Detection
High-resolution ROI inference for subtle object detection.

6

Long-Duration Videos
Efficient processing through motion-based selective inference.



RESEARCH AND REFERENCES

REFERENCE	WHAT IT ESTABLISHES	HOW VIGILENSE USES IT
Official Drishti AI Hackathon problem statement	Requirements, risks, hardware range, and human-oversight mandate	Primary scope and traceability baseline
NVIDIA DeepStream 8.0 documentation	GPU-accelerated streaming, inference, metadata, tracking, and plugins	Core offline video pipeline
DeepStream gst-nvof	Hardware optical flow from Turing/Jetson Orin onward; 4×4 block vectors	Primary motion-estimation layer
NVIDIA TensorRT support matrix	Hardware and precision compatibility	Prevents invalid FP4/engine portability claims
TensorRT-RTX precision guidance	FP4 is natively supported on Blackwell, not Ada	Basis for Gemma NVFP4 hardware decision
D-FINE official repository	ICLR 2025 detector, pretrained models, TensorRT/ONNX deployment	Primary detector candidate
RT-DETR/RT-DETRv2 official repository	Real-time DETR baseline, custom training, sliced inference, TensorRT	Detector challenger/fallback
MMPose	Pose framework including RTMPose and RTMO	Reproducible pose baseline
OC-SORT	Observation-centric tracking under occlusion/nonlinear motion	Tracker benchmark lineage
Deep OC-SORT	Appearance-assisted OC-SORT	A/B challenger to NvDCF
MMAction2	Pose/RGB temporal action-recognition framework	Optional temporal classifier evaluation
Roboflow Supervision	Detection structures, slicing, zones, annotations, evaluation utilities	UI overlays, crops, zones, metrics—not core intelligence
Google Gemma 4 12B model card	Native image/video understanding and model capabilities	Quality-mode visual verifier
Google Gemma 4 E4B	Mid-sized multimodal Gemma option with native image/video understanding	Default verifier family; use an instruction-tuned QAT Q4/GGUF build after runtime validation
Google Gemma 4 E2B QAT Q4 GGUF	Official lower-memory quantization-aware checkpoint	Constrained-GPU fallback verifier
AxionML Gemma-4-12B-NVFP4	Approximately 11 GB MLP-only NVFP4 checkpoint targeting Blackwell	Conditional Blackwell quality profile
NVIDIA Phi-4 multimodal NVFP4	NVIDIA-quantized 5.6B multimodal alternative	Optional NVIDIA runtime comparison
DVR-Scan	Classical motion-event detection for surveillance videos	External baseline for event segmentation
Cheatomaly	Temporal weakly supervised exam anomaly research	Supports temporal, not single-frame, evaluation direction
PySceneDetect	Shot/cut/fade detection	Explicitly rejected as the main activity segmenter